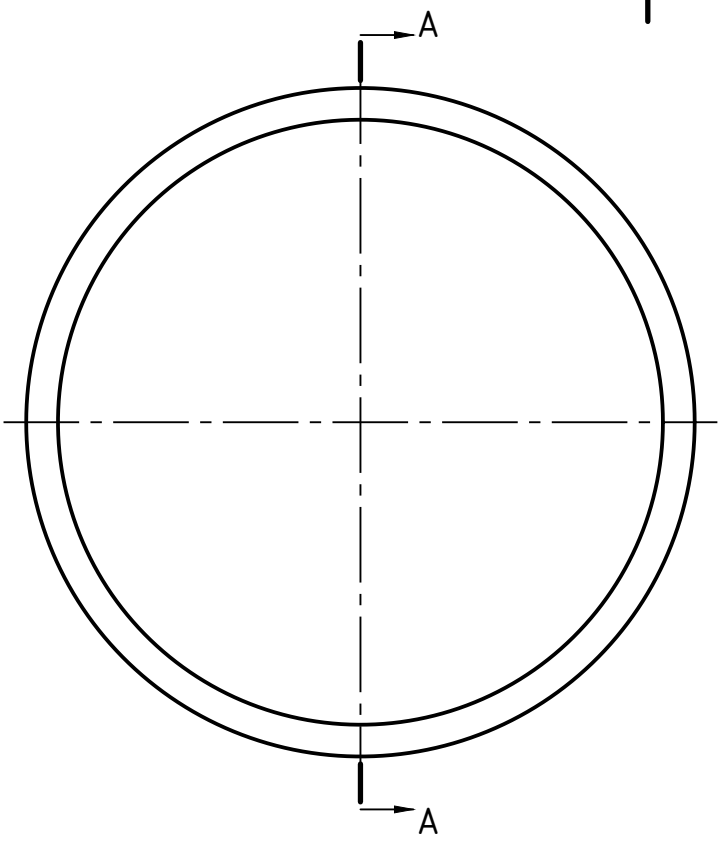
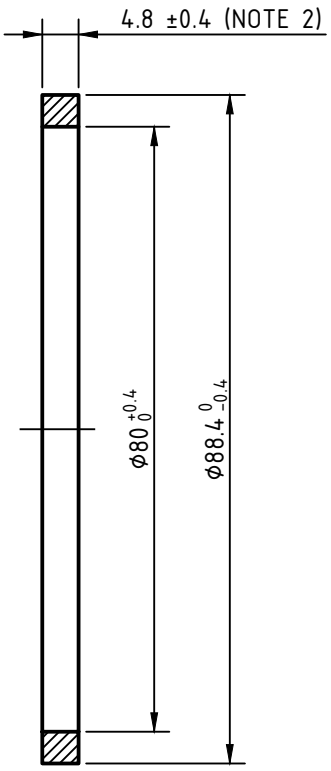
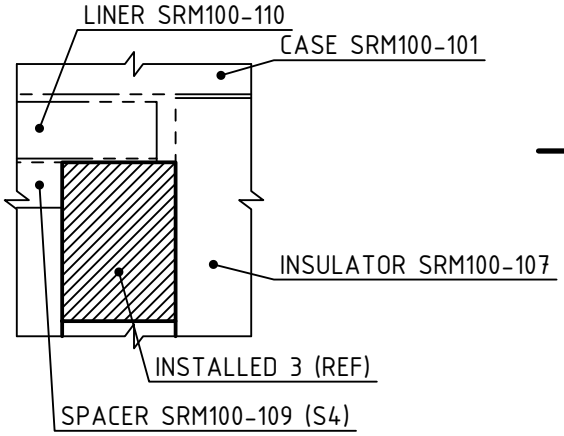




SECTION A–A

FREE STATE



NOTES

1. Material (certified): closed-cell silicone sponge, medium, ASTM D1056 2A2 / AMS 3195, e.g. Rogers BISCO HT-800; free density 0.30 – 0.40 g/cm³; compression set ≤ 10 % (22 h at 100 °C, 50 % deflection).
2. Stock: skinned 3/16 in sheet, free 4.8 ±0.4; where note 6 selects it, 3.2 or 6.4 ±0.4 of the same grade. Do not split or skive.
3. Cut to takeup_ring.dxf (1:1, mm, layer CUT) by steel rule die or oscillating knife; water jet acceptable; no laser or hot wire. The DXF sizes are limits (Ø88.4 max, Ø80 min): aim Ø88.2 / Ø80.2.
4. Inspect 100 %, no tears: free thickness t_s at 4 points 90° apart (dial gauge, flat foot, light load), mean on the bag; OD, ID by template, no squeeze.
5. Fit: OD in liner SRM100-110 bore Ø88.6 +0.1/0, diametral clearance 0.20 … 0.70 (free), whatever the case bore (Ø92.000 … 92.10). Bears on spacer SRM100-109 (S4) and insulator SRM100-107.
6. Take-up (SRM100-000 sheet 3): d_S from the case forward end face to the top of S4 and t from the closure flange face to the insulator's aft face, 4 points each, mean; g = d_S − t (nominal 3.00 ±0.50). Accept 0.50 ≤ g / t_s ≤ 0.75 (25–50 % compression), no shims: sheet 3.2 for g 1.60–2.40, 4.8 for g 2.40–3.60, 6.4 for g 3.20–4.80. Lay the ring on S4.

INSTALLED (5:1) – REFERENCE

7. ISO 10579-NR: free state, lying flat, at 23 °C. The 3D model and title-block mass are the ring installed, 3 thick: 3.33 cm³ × 0.50 g/cm³. As cut: 5.33 cm³ × 0.30 – 0.40 g/cm³ = 1.6 – 2.1 g.
8. Replace after every firing. Bag label: SRM100-108, batch, sheet, t_s; do not mark the part.
9. USED ON: SRM100-000 motor, item 12 (1 off); not fitted in SRM100-001.

MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
Silicone sponge, closed cell ASTM D1056 2A2, sheet 4.8	1.7 g	ALL TOLERANCED	–	SQUARE, AS CUT

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED – FOR REVIEW	1:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	TAKE-UP RING Grain-stack axial take-up – silicone sponge, cut from 4.8 sheet

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	–	2026-09-23	SRM100-108	A	1 / 1